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⇒ ANALYSIS OF ALGORITHMS

⇒ Assignment - 1

Q1)

1. Algorithmic Strategy.

The algorithm uses a single traversal of the array.
First, the first element is assigned as both the largest and smallest element.
then, the remaining elements are checked one by one.
if an element is greater than the current largest value, the largest value is updated.
if an element is smaller then the current

smallest value, the smallest value is updated.

After checking all elements, the algorithm returns the largest and smallest values.

2. Pseudocode

```
Algorithm FindLargestSmallest(A, n)
  Largest ← A[0]
  Smallest ← A[0]
  for i ← 1 to n-1 do
    if A[i] > Largest then
      Largest ← A[i]
  if A[i] < Smallest then
    Smallest ← A[i]
return Largest, Smallest
```

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3. Major Primitive Operations

The major primitive operations are:

- Assignment
- Comparison
- Array indexing
- Increment of the loop variable
- Return operation

The main comparisons are:

A[i] > Largest

A[i] < Smallest

4. Best-Case and Worst-Case Running Time.

The algorithm must examine every element of the array in both cases

for an array of n element, the loop runs n-1 times.

Best Case:

$T_{best}(n) = c(n-1) + c$

Therefore:

Best Case = $\Theta(n)$

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Worst Case:

$T_{worst}(n) = c(n-1) + c$.

Therefore:

worst case = $\Theta(n)$.

5- Asymptotic Notation

the running time grows Linealy with the input size.

- $O(n)$
- $\Omega(n)$
- $\Theta(n)$

Therefore, the overall time Complexity is:

- $\Theta(n)$

6- Space Complexity

The algorithm uses only a Constant number of additional variable.

- largest
- Smallest
- i

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No additional array is created.
Therefore:
Space Complexity = O(1).

Q2)

Algorithm 1 : Linear Search.

7. Strategy.

Liner search checks the elements one by one form the beginning of the array until the valy X is found or the end of the array is reached.

8. Pseudocode.

```
Algorithm LinearSearch (A, n, X)
for i ← 0 to n-1 do
    if A[i] = X then
        return TRUE
return FALSE
```


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9. Best-Case and Worst-Case Time Complexity

Best Case:

if X is the first element, only one comparison is required.

- Best case = $\Theta(1)$

Worst Case:

if X is the last element or does not exist, all n element must be checked.

- Worst Case = $\Theta(n)$.

10. Asymptotic Notation

For the best case:

- $O(1)$
- $\Omega(1)$
- $\Theta(1)$

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For the worst case:

- $O(\log n)$
- $\Omega(\log n)$
- $\Theta(\log n)$

Overall worst-case Complexity:

- $\Theta(\log n)$.

II. Comparison of the Two Algorithms

Feature	linear search	Binary search
Array condition	Can be unsorted	Must be sorted
Best case	$\Theta(1)$	$\Theta(1)$
Worst case	$\Theta(n)$	$\Theta(\log n)$
Extra Space	$O(1)$	$O(1)$
Strategy	Check element on by one	Divide the search range by two

Comparison :

Liner Search is more suitable when the array is unsorted, the input size is small, or only a few searches are required.

Binary Search is more effient for large arrays when the data is already sorted and many searches are required

12- Effect of Input Size and Arrangement

The choice of algorithm depend on the size and arrangement of the input data.

For a small or unsorted array, Linear search is simple and suitable because it does not require the array to be step

Therefor :

small / Unsorted Data → Linner Search

Large / Sorted Data → Binanary search

Final Counclusion :

Linear Search → $\Theta(n)$

Binary Search → $\Theta(\log n)$

For large sorted data, Binary Search is more efficient